

After TradeLens: A Permissionless Bonded Replacement

Alessandro Daliana

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Abstract

TradeLens (2018–2023), the IBM–Maersk container-shipping consortium that aimed to produce a tamper-proof shared record of every container’s journey, shut down with only one major non-Maersk carrier on board. The shutdown was not a software failure. It was a governance failure visible from the consortium’s structural shape: a competitor-controlled platform, requiring participation from carriers who would not ratify a competitor’s gate-keeping over an industry resource, with high integration cost and no settlement guarantee binding parties to the arrangement. The failure was structural, predictable from the shape of the architecture, and not specific to TradeLens.

The architectural alternative we develop operates at the same perimeter TradeLens attempted: inter-logistics coordination among ocean carriers, ports of loading and discharge, customs authorities, freight forwarders, NVOCCs, customs agents, trade-finance parties / LC issuers, marine insurers, and inspection services. We present a permissionless, ownerless bonded composition built on the Figaro kernel’s two mechanisms (asymmetric bonding plus buyer dominance with atomic resolution). The composition has no consortium, no foundation, no admin function, no upgrade path, and no central party who can be ratified or refused. Each provider is an off-chain real-world asset (RWA) whose on-chain participation is mediated by a wallet, with a human or autonomous agent operating the wallet on the asset’s behalf. The `rootBuyer` at this perimeter is the buyer-of-record at the relevant tier — typically the importer-of-record, the forwarder-of-record, or the consignee depending on the Incoterm and the operational arrangement; the kernel is agnostic about which party occupies the buyer slot. Bonded commitments at each commit lock skin in the game proportional to the cumulative process value; atomic resolution by the buyer-of-record settles the whole process on receipt of the goods at the agreed delivery point. Carriers who want to participate compose; carriers who do not do not, with no consequence for either decision.

The schemas the composition draws on are largely existing infrastructure (commerce, geo, handoff, jurisdiction, fulfilment, GHG family, proximity); two new schemas (container-seal, Incoterms-2020-as-anchor) extend the protocol’s surface in ways that respect the extension-doctrine. We also treat the bill-of-lading transferability question through the cancellable-seller / counter-process pattern that expresses transferability at the protocol layer without weakening the kernel’s invariants.

The paper is in industrial-engineering register: the contribution is the composition and its operational properties, not novel mechanism-design results. Why TradeLens failed and what a bonded replacement at the same perimeter looks like architecturally are answerable as supply-chain coordination questions.

Keywords: container shipping, supply-chain coordination, bills of lading, Incoterms, MLETR, transferable records, process modeling

1 Introduction

Container shipping is a coordination problem of unusual scale: a single 20,000-TEU vessel carries cargo from thousands of shippers, moves through multiple ports under multiple jurisdictions,

transfers to multiple consignees, and is documented by a stack of bills of lading, customs filings, GHG attestations, and operational handoffs that no single party originates and no single party owns. The information problem is real — duplicated paper-based records, opaque vessel-position data, contested seal-integrity claims — and the industry has been searching for a digital solution for two decades.

TradeLens, the consortium platform jointly developed by IBM and Maersk between 2018 and 2023, was the most ambitious attempt at that solution. Its approach was a permissioned blockchain on which participants would publish operational records that all parties to a shipment could read. The platform shut down in early 2023 after failing to attract more than one major non-Maersk carrier (CMA CGM) into participation; MSC, Hapag-Lloyd, ONE, and Evergreen all declined. The platform was technically functional; the consortium was not.

Reading the failure structurally. The standard reading of TradeLens’s failure focuses on adoption dynamics: the network-effects argument that any shared-ledger platform requires a critical-mass coalition before it is useful. This is not wrong, but it is incomplete. The critical-mass argument explains why TradeLens *failed to grow*; it does not explain why the carriers who declined to join made what was, from their perspective, a rational decision. The carriers’ decision was not about technology adoption. It was about ratifying a competitor’s control of an industry resource, which a competitor-controlled platform necessarily asks them to do.

The *governance* structure of the platform was the binding constraint, not the network effects. A competitor-controlled platform requires the other competitors to accept the gatekeeping authority of the controlling competitor; the other competitors have correctly read this as conceding strategic position and have declined. The network-effects collapse follows from the governance choice; it does not cause it. A platform with neutral governance might have faced its own difficulties, but it would not have faced the specific difficulty TradeLens did.

The architectural alternative. The structural fix is not better governance but no governance at the platform layer at all. A permissionless, ownerless settlement primitive — the Figaro kernel’s bonded commitment — removes the governance question from the architecture. There is no consortium to join; there is a public protocol that any wallet can compose with. There is no gatekeeping authority to refuse; carriers participate by signing commitments and decline by not signing. The network-effects question becomes more manageable because the platform-as-veto problem disappears: a competitor’s participation does not require the competitor’s consent to a competitor’s gatekeeping.

Wallets, real-world assets, and the perimeter. Every party in the bonded composition is a wallet representing an off-chain real-world asset (RWA) with productive, commercial value — an ocean-going vessel, a port terminal’s berth and crane capacity, a customs authority’s clearance service, a forwarder’s multi-modal coordination capability, an inspection service’s on-the-ground presence, a marine insurer’s underwriting capacity. The wallet holds the asset’s earnings as token balances and points to the asset’s off-chain credentials through NFTs (a vessel’s classification certificate, a port operator’s authority licence, a customs broker’s customs licence, an insurer’s policy authorisation). A human or autonomous agent operates the wallet on the asset’s behalf. Public-authority wallets (port authorities, customs authorities) participate on the same footing as commercial providers; the kernel does not distinguish a tax from a fee for service. The full RWA-as-wallet apparatus is developed in the accounting paper of this corpus.

The perimeter the paper develops. This paper develops the bonded composition at the same perimeter TradeLens attempted: *inter-logistics coordination* among the parties that move a container from origin to destination — ocean carriers, ports of loading and discharge, customs authorities, freight forwarders, NVOCCs, customs agents, trade-finance parties / letter-of-credit

issuers, marine insurers, inspection services, and cold-chain monitors. The `rootBuyer` at this perimeter is the buyer-of-record at the relevant tier (importer-of-record, forwarder-of-record, or consignee, depending on the Incoterm and operational arrangement); the kernel is agnostic about which party occupies the buyer slot. Granularity within the perimeter is a design choice the contracting parties make, not a kernel constraint — the same movement of goods can be composed as a coarser process (a few commits to aggregator wallets) or a finer one (more commits to specialised value-adders), with the bonding equilibrium holding at every edge regardless. The composition is not a TradeLens replacement in the sense of a drop-in alternative product; it is a re-architecture of the inter-logistics coordination problem in which the consortium-platform layer dissolves into a permissionless settlement primitive.

Paper organization. Section 2 reviews the TradeLens history and the structural reading of its failure. Section 3 summarizes the settlement primitive in the register the present paper uses. Section 4 develops the inter-logistics composition: the organizational DAG (origin to destination through the inter-logistics value-adders), the schema-attested operational handoffs along it, and the kernel commit structure that backs it. Section 5 works out the bond posture and cohort-compensation mechanics across the resource providers, with a worked example. Section 6 treats the schema requirements: most existing infrastructure, two new schemas (container-seal-v1 and Incoterms-2020-as-anchor), and the deliberate non-inclusion of transferable-BoL semantics in the kernel-layer schema set. Section 7 treats bill-of-lading transferability as a protocol-layer composition pattern (CancellableSeller wrapper + counter-process). Section 8 compares the architecture to CargoX, TradeTrust, MLETR, and TradeLens itself. Section 9 states the scope conditions and what the architecture deliberately does not address. Section 10 concludes.

2 What TradeLens Was and Why It Failed

We treat the failure structurally rather than narratively.

What TradeLens was. TradeLens was a permissioned blockchain platform for global container shipping, jointly developed by IBM and Maersk and operational from 2018 to 2023. The platform’s stated goal was a shared, tamper-proof record of every container’s journey — bills of lading, customs filings, port handoffs, vessel positions — accessible to all parties in a shipment.

The technical architecture was Hyperledger-Fabric-based: a permissioned ledger where participating carriers, ports, and customs authorities could publish operational records. The platform’s value proposition rested on the assumption that all major ocean carriers would join, producing a single ledger that shippers could rely on as comprehensive cross-carrier reference. Jensen et al. [2019] and Sarker et al. [2021] document the platform’s value proposition and use-case design as it stood mid-deployment; both papers predate the shutdown and frame the platform around adoption potential rather than around the governance refusal we read as the binding constraint below. The governance-was-binding reading is original to this paper; Jensen and Sarker supply the value-proposition baseline against which the reading is offered.

The adoption shape. By 2022, only CMA CGM (a Maersk competitor) had joined as a major non-Maersk carrier. MSC (the largest carrier by capacity), Hapag-Lloyd, Ocean Network Express (ONE), and Evergreen had all declined to participate. In November 2022, IBM and Maersk announced the platform’s shutdown; production ended in Q1 2023 [Lloyd’s Loadstar, 2022].

Why the carriers declined. The standard adoption-dynamics reading explains the shutdown through a familiar mechanism: shippers want a single ledger covering 90%+ of their cargo

volume; without MSC and Hapag-Lloyd the data was incomplete; without complete data, shippers wouldn't pay for the platform; without revenue, carriers had no incremental reason to join; the equilibrium was reflexive non-adoption. This is correct as far as it goes.

The deeper reading is that the non-Maersk carriers correctly identified a strategic problem with the platform's governance shape. A platform controlled by Maersk, with IBM as the technical operator and Maersk as the principal customer, is structurally a Maersk asset that other carriers are being asked to provision with their operational data. The asset's governance accountability runs to Maersk in any contested case. Whatever the platform's formal governance arrangements, the controlling-competitor dynamic is the binding constraint on competitors' participation.

The argument is not that Maersk acted in bad faith or that the formal governance was inadequate. The argument is that the platform's shape *is* the problem: a competitor-controlled infrastructure requires other competitors to ratify the controller's strategic position relative to the industry, and there is no formal-governance fix that dissolves this requirement. MSC and Hapag-Lloyd are not joining a Maersk-controlled platform regardless of how its governance is configured.

The general lesson. TradeLens's failure is not specific to IBM, to Maersk, or to shipping. It is the recurring pattern of consortium-governance infrastructure: any platform whose governance accountability runs to a participant of the industry the platform is meant to serve will face structural participation refusal from that participant's competitors. The fix is not better consortium governance; the fix is no consortium at the platform layer.

Several adjacent industries have run the same experiment with the same outcome: SWIFT and its alternatives in payments, various consortium attempts at securities settlement, healthcare-records consortia. The pattern is robust. The structural alternative is permissionless, ownerless infrastructure at the platform layer, with whatever industry-specific arrangements the participants want to compose on top.

3 The Settlement Primitive

The bonded primitive in FigaroCore has two mechanisms. We summarize the features the present argument uses.

Asymmetric bonding (Mechanism 1). For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. Cooperation is the unique profile surviving iterated elimination of weakly dominated strategies. The mechanism scales to N -party processes through progressive collateralization: each seller bonds against the cumulative value of all upstream commitments, producing a mesh of independently secured bilateral edges, each carrying its own equilibrium.

Buyer dominance with atomic resolution (Mechanism 2). Only the root buyer can trigger resolution; resolution settles all active orders in the process simultaneously or not at all. The all-or-nothing rule induces a weakest-link subgame among sellers, producing cooperation pressure of magnitude $P_i + 2G_i$ on every other seller from each seller's perspective, without explicit communication, governance, or managerial layer.

What this implies for shipping coordination. Mechanism 1 prices each leg's enforcement in proportional bond posture without requiring any consortium or platform to underwrite it. Mechanism 2 enforces the multi-leg coordination at resolution without requiring any party to

govern the multi-leg arrangement. The two mechanisms together supply the coordination function that TradeLens attempted to supply through a consortium-governed shared ledger; they supply it without needing the consortium.

4 The Assembly

We now describe the bonded composition that operates at the TradeLens perimeter. The composition has two layers that must be kept distinct: the *organizational DAG* (how the container physically moves from origin to destination through the inter-logistics value-adders, documented via schema attestations) and the *kernel commit structure* (who commits to whom on-chain, governed by the kernel's `buyer = rootBuyer` rule). The kernel is DAG-blind; the schema layer is DAG-aware. The two layers co-exist in one process.

The organizational DAG. A container shipment is a sequence of operational handoffs from origin to destination. We sketch a canonical international flow, with the inter-logistics value-adders made explicit:

```

shipper-of-record (origin country)
  |
  | handoff: goods presented for shipment
  v
freight forwarder + NVOCC
  | (forwarder coordinates multimodal
  | movement; NVOCC issues house BoL)
  v
origin inland leg (truck or rail to port)
  |
  | handoff: container stuffed and sealed
  v
inspection service (pre-shipment, optional)
  |
  | container-seal-v1 inspection
  v
port-of-loading (terminal services)
  |
  | handoff: container loaded; master BoL issued
  v
ocean carrier
  |
  | vessel-position stream (geo-v1),
  | container-seal-v1 sequence,
  | cold-chain monitor (if reefer)
  v
port-of-discharge (terminal services)
  |
  | handoff: container discharged
  v
customs agent + customs authority
  | (agent files entries on behalf of
  | importer; authority bonded for
  | clearance service and tariff)

```

```

      v
destination inland leg
      |
      | fulfilment-v1 (delivery to consignee)
      v
consignee (importer-of-record / buyer-at-tier)

```

Marine insurance and trade-finance / letter-of-credit issuance operate as parallel value-adders bonded against the same process without sitting on the cargo-flow line — the marine insurer’s wallet bonds against the goods’ value, the LC issuer’s wallet bonds against the payment-on-documents arrangement, and both are sellers in the bonded composition without owning a physical handoff. Variations on the structure (intermodal rail substituted for inland truck, multiple consolidation points, transshipment ports, etc.) instantiate the same pattern with the operational divisions shifted to reflect the route. Every handoff along this DAG is documented via a schema-typed attestation: `handoff-v1` for cargo transfers, `container-seal-v1` for seal-state changes, `geo-v1` for vessel-position updates, `jurisdiction-v1` for customs clearance, `fulfilment-v1` for delivery confirmation. None of these attestations is a kernel commit; they are evidentiary records documenting the operational reality of the container’s flow.

The kernel commit structure. The kernel commits, by contrast, all share one buyer. The kernel enforces `buyer = rootBuyer` in every order in a process; the `rootBuyer` at this perimeter is the *buyer-of-record at the relevant tier*. Depending on the Incoterm and the operational arrangement, that party can be the importer-of-record, the consignee, a forwarder-of-record acting on the importer’s behalf, or in seller-arranged shipments the shipper-of-record itself. The kernel is agnostic about which party occupies the buyer slot. We use *importer-of-record* as the running illustration; the apparatus is the same with any of the alternatives.

The commits document who the importer-of-record pays and how much: `importer → shipper-of-record` (for the cargo’s invoice value, if the Incoterm assigns the import side that obligation), `importer → freight forwarder`, `importer → NVOCC`, `importer → origin inland carrier`, `importer → inspection service`, `importer → port-of-loading authority`, `importer → ocean carrier`, `importer → port-of-discharge authority`, `importer → customs agent`, `importer → customs authority` (for the import duty), `importer → destination inland carrier`, `importer → marine insurer` (if buyer-procured cover), `importer → LC issuer` (if the payment is letter-of-credit-arranged), and any other resource provider the operational arrangement draws on. The commits are independent of who hands the container to whom in the organizational DAG above. The importer alone calls `resolveProcess` on receipt at the agreed delivery point, settling the process atomically.

Granularity is a design choice. The same shipment can be composed coarsely (a few commits to aggregator wallets — the forwarder absorbs origin inland + port-of-loading + carrier + port-of-discharge + destination inland into one bonded commit) or finely (each value-adder a separate commit, as enumerated above). The kernel sees only commits; it does not reify any tier-role or aggregation choice. Granularity is chosen by the contracting parties to suit their coordination needs, with the bonding equilibrium holding at every edge regardless. This paper develops the fine-grained version because the value-adder structure is the substantive content of the TradeLens-perimeter coordination problem; coarse aggregations are admissible and may be operationally appropriate in many deployments.

Off-chain organizational shapes. The conventional inter-logistics chain involves named organizational shapes (shipper-of-record, consignee, forwarder, NVOCC, customs broker, importer-of-record, customs agent). Under the bonded composition the kernel does not see these tier-roles;

each tier-role is whichever wallet the contracting parties designate at agreement signing. Off-chain organizational shapes that today perform discovery, curation, brand-as-quality-signal, or coordination services around the inter-logistics layer can still be filled by parties providing those specific services, but they cannot insert themselves as kernel intermediaries between the buyer-of-record and the operational sellers; the kernel forbids any party other than the rootBuyer from being the buyer of within-process commits.

Per-commit schema clauses. Each kernel commit carries the schema clauses relevant to the seller’s role.

Each commit pairs the standard `commerce-v1` payment clause with the schema clauses appropriate to the seller’s role.

- *Shipper-of-record*: `handoff-v1` on goods presented for shipment, optional `ghg-protocol-v1` disclosure for manufacturing emissions.
- *Freight forwarder*: `handoff-v1` on cargo receipt at consolidation; `container-seal-v1` applied at the consolidation point.
- *NVOCC*: house bill-of-lading reference (the BoL itself is treated as a protocol-layer composition pattern in Section 7).
- *Origin and destination inland carriers*: `handoff-v1` at pickup and dropoff; `geo-v1` pickup and dropoff geohashes.
- *Inspection service*: `container-seal-v1` inspection event; optional evidence-hash pointer to a pre-shipment inspection report.
- *Port-of-loading and port-of-discharge authorities*: payment for terminal handling and gate access; `handoff-v1` at port-side transfers; `container-seal-v1` inspected-intact attestations at each port handoff.
- *Ocean carrier*: an `incoterms-2020-v1` clause anchoring the shipment term and named place; `handoff-v1` on transfer to the vessel and at discharge; `container-seal-v1` sealing events through the voyage; `geo-v1` vessel-position attestations on the voyage cadence; `ghg-protocol-v1` disclosure for voyage emissions; for reefer cargo, cold-chain attestations.
- *Customs agent*: `jurisdiction-v1` attestation recording the customs authority’s determination, plus payment for entry-filing services.
- *Customs authority*: payment for the import duty. The customs-authority wallet is an RWA-as-wallet of the same kind as a commercial provider — a public service sustained by receipts covering its operating expenses — in the role of a bonded seller of border-clearance service. The sovereign decisional authority on the customs agent’s parallel attestation is an off-chain regulatory function the kernel does not encode.
- *Marine insurer*: payment of the policy premium; the insurer’s bond is the underwriting capacity at risk against the insured cargo value. Insurance composes with the bonded primitive without modifying the kernel.
- *LC issuer* (if applicable): payment of the LC issuance fee; the issuer’s bond is the credit capacity backing the documentary-payment arrangement.

Operational sub-procurement as separate processes. Each value-adder’s own procurement (the carrier’s fuel + crew + maintenance, the forwarder’s IT + staffing, the inspection service’s labour + equipment, the port authority’s terminal operations, etc.) is a separate process under that wallet’s operator as its own `rootBuyer`. The kernel does not link those processes to the importer’s. Cross-process coordination is achieved through within-process cohort-pressure on each side and through off-chain reconciliation, not through cross-process atomic resolution. This is the standard wallet-as-RWA composition pattern: each provider wallet sustains its participation in this market through receipts that cover the asset’s off-chain operating expenses, and runs its own bonded processes for those expenses against its own counterparties.

5 Bond Posture and Cascade Mechanics

The composition’s structural contribution is that each leg’s risk is priced and propagated by the bonding rule of the kernel, without consortium intermediation.

Bond posture per commit. The kernel maintains a single process-wide cumulative-value accumulator G that grows monotonically as each commit lands ($G \leftarrow G + P_i$). At every commit the importer-of-record (as `rootBuyer`) posts $2P_i$ for that specific payment, and the seller posts $2G_{\text{at commit time}}$ — the full cumulative process value at the moment they commit. Bond posture is asymmetric: the importer’s per-commit bond scales with the immediate payment, while each seller’s bond scales with the cumulative value of every upstream commit in the process. Sellers committing later post correspondingly larger bonds; the last seller bonds against the full landed value of the cargo.

A bond-posture worked example. Consider a stylized \$1,000 cargo invoice (containerised goods shipped from an exporting country to an importing country with a 5% import duty assessed on the CIF declared value), with the importer-of-record as `rootBuyer`. The table below traces one illustrative commit sequence.

The asymmetry between importer-side and seller-cohort bonds is visible: the importer’s aggregate locked bond is \$2,000 ($= 2 \times P_{\text{cargo invoice}}$, the standard buyer posture), while the seller cohort’s aggregate locked bond is \$21,886, nearly eleven times the importer’s. The seller cohort’s bond is a large multiple of the cargo invoice because each seller bonds against the cumulative G that has grown across upstream commits, and the inter-logistics perimeter has a substantial number of value-adders. As the composition adds more value-adders (e.g., transshipment ports, additional inland legs, multiple inspection services) the seller-side total grows super-linearly while the importer’s side stays at $2P_{\text{cargo invoice}}$.

Slip propagation as cohort-compensation pressure. When a slip occurs — a damaged container, a port congestion delay, a customs hold, a documentary discrepancy, a cold-chain breach — the kernel’s atomic-resolution rule means *every* seller’s bond in the importer’s process is at risk: the importer alone resolves, and the importer can withhold resolution while the seller cohort negotiates compensation. The unit whose performance failed bears the larger share of cohort-internal compensation; sellers whose service was unaffected contribute less or not at all. Compensation flows directly from the seller cohort to the importer before any external mechanism (specialized mechanism contracts, dispute forums, insurance, regulators) is engaged. The mechanism is buyer dominance plus atomic resolution under bond pressure: cooperation is the unique profile surviving iterated elimination of weakly dominated strategies, induced by atomic resolution operating on the bonded mesh. The same structural shape produces the joint-liability dynamic in Grameen-style microfinance, but operating without the social-substrate prerequisites Grameen requires (no repeated interaction, no local information, no exogenous punishment technology). Buyer dominance with atomic resolution is in this sense a *social mechanism*: it produces

Seller (commit sequence)	P_i	G after commit	Importer bond $2P_i$	Seller bond $2G$
Shipper-of-record (cargo, root)	\$700	\$700	\$1,400	\$1,400
Marine insurer	\$15	\$715	\$30	\$1,430
Freight forwarder	\$30	\$745	\$60	\$1,490
NVOCC (house BoL)	\$15	\$760	\$30	\$1,520
Origin inland carrier	\$20	\$780	\$40	\$1,560
Inspection service (pre-shipment)	\$10	\$790	\$20	\$1,580
Port-of-loading authority	\$25	\$815	\$50	\$1,630
Ocean carrier	\$60	\$875	\$120	\$1,750
Port-of-discharge authority	\$25	\$900	\$50	\$1,800
Customs agent	\$20	\$920	\$40	\$1,840
Customs authority (import duty)	\$39	\$959	\$78	\$1,918
Destination inland carrier	\$25	\$984	\$50	\$1,968
LC issuer (documentary credit fee)	\$16	\$1,000	\$32	\$2,000
Total bond locked across the process	—	—	\$2,000	\$21,886

Table 1: Stylized per-commit bond posture for a \$1,000 cargo shipment under one TradeLens-perimeter process with the importer-of-record as `rootBuyer`. The import duty (\$39) is 5% of the declared CIF value (\$775 = \$700 cargo + \$15 marine insurance + \$60 ocean freight). G is the process-wide cumulative-value accumulator, monotonically growing with each commit. The importer’s bond at each commit scales with the immediate payment; each seller’s bond scales with the cumulative process value at the moment that seller commits. The customs-authority wallet is an RWA-as-wallet on the same footing as commercial providers — a public service whose continued operation depends on receipts covering its operating expenses — in the role of a bonded seller of border-clearance service. Per-leg payments are illustrative; absolute magnitudes vary with cargo class, route, tariff schedule, and the operational arrangement (open-account vs. documentary-credit, FOB vs. CIF, buyer-procured vs. seller-procured insurance, etc.).

social-coordination behavior (peer pressure, cohort negotiation, burden-sharing, coverage of a struggling counterpart) endogenously through the bond architecture, with each provider wallet’s continued participation in its market making the negotiation rationally compelled rather than socially expected. The off-chain question of which operational unit caused the slip — forwarder coordination failure, terminal damage, ocean-leg damage, customs inspection delay, documentary discrepancy at the LC stage — is adjudicated by external forums with the bonded record as input.

Cohort pressure across the resource provider set. The composition exhibits cohort pressure across the seller set in the importer’s process. Each seller’s bond is at risk if any other seller’s slip propagates to a resolution-withholding by the importer. Sellers therefore have direct economic interest in each other’s reliability: a forwarder whose coordination failures force repeat compensation negotiations loses repeat importer-of-record business not through reputation effects but because the cohort’s expected bond-loss exposure on shipments through that forwarder is direct and measurable; a port whose congestion consistently produces cascading delays does likewise. The mechanism operates without a managerial layer; the bonded structure does the work that a consortium-governed shared ledger was meant to do through information transparency alone.

6 Schemas

The composition draws mostly on existing schema infrastructure. Two new schemas are required: a container-seal attestation and an Incoterms-2020 anchor. The bill-of-lading transferability

question is treated separately in Section 7.

Existing schemas. The composition reuses the following.

- **figaro-commerce-v1** — per-leg payment commitment.
- **figaro-geo-v1** — pickup / drop-off geohashes and vessel-position updates.
- **figaro-handoff-v1** — each container handoff at transfer points.
- **figaro-jurisdiction-v1** — customs-clearance attestation.
- **figaro-courier-process-v1** — the carrier’s per-role event log over the multi-stop carrier process.
- **figaro-fulfilment-v1** — final delivery to the consignee.
- **figaro-ghg-protocol-v1** (with sister schemas for ISO 14064, PAS 2050, EN 16258) — per-leg emissions disclosure on every transport edge.
- **figaro-proximity-policy-v1** together with **figaro-proximity-proof-v1** — geofence enforcement at port handoffs where required.

These schemas are already on chain and bound to validators; no further protocol-extension work is needed for them.

New schema: figaro-container-seal-v1. Container seals are the integrity mechanism for ocean shipping. A uniquely-numbered seal is applied at the origin and broken only at customs or at the destination consignee. Multiple parties (carrier, ports, customs, consignee) need a single canonical attestation of when the seal was applied, when it was inspected intact, and if it was breached.

The schema’s Layer A spec carries:

- **containerNumber** (string, ISO 6346 format)
- **sealNumber** (string)
- **event** (enum: applied, inspected_intact, transferred, breached, removed_by_customs)
- **inspectorAddress** (address-hex; the wallet making the attestation)
- **timestamp** (ISO 8601 datetime)
- **locationGeohash** (string, 8-character precision)
- **evidenceHash** (bytes-hex, optional content hash of a photo or inspection report)

The schema is the seed of a *container-integrity* family that could later cover seal-equivalents in other transport modes (rail container seals, air-freight tamper-evident packaging, etc.) without reauthoring the family. The `schemaId` binds permanently under the `SchemaRegistry`’s first-write-wins discipline.

New schema: figaro-incoterms-2020-v1. Incoterms 2020 (from the International Chamber of Commerce) are the international trade vocabulary for risk-transfer points and cost allocation across legs of a shipment: EXW, FCA, CPT, CIP, DAP, DPU, DDP, FAS, FOB, CFR, CIF [ICC, 2020, Ramberg, 2010]. The vocabulary is useful as cross-party shared reference, but the traditional Incoterms semantics come from a context without the bonded primitive’s invariants and may not all map cleanly. We treat this carefully because it is the canonical worked example of how traditional commercial vocabulary smuggles in assumptions the bonded architecture cannot honor.

Three structural mismatches deserve direct treatment.

Risk transfer is a Figaro-foreign concept. Incoterms encode a discretionary risk-transfer point: “risk transfers from seller to buyer at point X.” Under the bonded primitive, risk follows bond — the bonded party is at stake until they perform per the agreed clauses, and there is no separate *risk-transfer* concept distinct from the bonding rule. A clean port from Incoterms to bonded clauses requires reinterpreting “risk transfer at X” as “the seller’s **handoff-v1** attestation due at X discharges the seller’s bond exposure for the upstream segment.”

Cost-vs-risk separation collapses. Incoterms separate “who pays for transport” from “who bears risk during transport.” Under the bonded primitive, both follow from process structure: a party who bonds for a sub-process is the buyer of that sub-process, and risk is what the bonding rule already prices. Some Incoterms features (CIP/CIF’s seller-pays-insurance feature, for instance) require composition with a parallel insurance process rather than direct encoding in the delivery-clause specification.

Buyer dominance changes the resolution model. Incoterms assume good-faith resolution between parties; the bonded primitive encodes buyer dominance plus mutually-assured bond-loss as the equilibrium that produces good-faith resolution. Some Incoterms’ implicit dispute-resolution paths — which typically rely on local commercial-court adjudication — continue to function under the bonded architecture but at the off-chain forum the parties name in their agreement, not as an internal property of the term.

The right schema is therefore not “Incoterm as a traditional contract clause” but *Incoterm as a reference to a Figaro-native delivery-clause specification*. The schema anchors the term plus the named place; the term-to-delivery-clause mapping lives in the off-chain ICC publication (the canonical text) and a runtime term-to-clause table that the assembly’s participants compose from. The validator contract enforces the anchor (term enum, named place string, place geohash for cross-reference with **geo-v1**) but does not encode per-term semantics in Solidity.

The Layer A spec carries:

- **term** (enum: EXW, FCA, CPT, CIP, DAP, DPU, DDP, FAS, FOB, CFR, CIF)
- **namedPlace** (string; every Incoterm requires a named place)
- **placeGeohash** (string, 8-character precision for cross-reference with **figaro-geo-v1**)

This split lets future Incoterms revisions ship as new schemaIds (**figaro-incoterms-2030-v1**, etc.) without mutating prior anchors. Per-term mapping verification — which terms map directly, which require composition with parallel processes (insurance, customs), and which do not transfer at all — is work the schema-author performs against the kernel code before the schema is registered. Likely outcomes from that verification are the following.

EXW, FCA, DAP, DPU, DDP, FAS, FOB probably map directly, each becoming a delivery-clause specification: **handoff-v1** attestation at the named place, with auxiliary clauses for customs (DDP) or unloading (DPU) where the term requires them.

CPT and CFR map structurally. The “seller pays carriage” feature becomes a separate sub-process where the seller is the buyer of carriage. The “risk transfers at first carrier” feature becomes the goods-commit’s delivery-clause specification.

CIP and CIF map similarly to CPT and CFR plus the “seller insures for buyer’s benefit” feature. The insurance assignment may not transfer directly — it likely requires composition with a parallel insurance process where the seller buys insurance from an external insurer naming the buyer as the beneficiary. The schema-author’s verification report identifies this as a composition requirement rather than as a refusal.

The Incoterms schema is therefore a clean illustration of the extension-doctrine principle that the protocol anchors shared references but does not import every contractual assumption that travels with traditional vocabulary. The vocabulary is useful; the assumptions are sometimes incompatible; the schema separates them.

Schemas the assembly does not include. The assembly does not include a multi-currency-cross-leg schema (would break same-unit comparability and the bonding equilibrium), a force-majeure-escape schema (would re-introduce the kind of unilateral exit the kernel forbids), or a transferability schema internal to `commerce-v1`. The transferability question is treated separately in Section 7 as a protocol-layer composition pattern.

7 Bill of Lading Transferability

The bill of lading is the carrier’s receipt of cargo, the contract of carriage, and (in the negotiable case) a document of title. The transferability question — whether the right to receive cargo at destination can be transferred mid-shipment from the original consignee to a new consignee — is doctrinally significant because container-shipping commerce often involves cargo sale in transit, and the canonical legal artifact for that sale is endorsement of the bill of lading.

The bonded primitive’s kernel does not support party substitution within a single bonded order. Three kernel invariants each independently rule it out: the buyer is fixed at root commitment and identical across every sub-order in the process; the parties to each commitment are fixed at the moment of commit; and the no-escape-hatches discipline forbids unilateral exit paths from a bonded state. A literal endorsement of the BoL during shipment would require any of these invariants to bend, and they do not bend.

The architectural direction is that BoL transferability is a *protocol-layer composition*, not a kernel-layer property. The candidate pattern is a *cancellable-seller wrapper plus counter-process*: the original buyer, who wants to transfer the right to the cargo mid-shipment, commits a parallel *cancellation process* P_{cancel} in which each sub-seller is compensated for early exit through a fee owed by the original buyer (and bonded under the standard kernel rule), and the new buyer commits a fresh transport process for the remaining legs.

Axes of underspecification. The pattern is sketched here as a candidate, not asserted as a deliverable mechanism. It is underspecified along three axes that bound how seriously a deployment can rely on it.

Sub-seller consent under the wrapper. The cancellable-seller wrapper relies on programmatic acknowledgement of the cancellation by the affected sub-seller under a pre-agreed fee schedule. Whether this constitutes sub-seller consent in the kernel-discipline sense, or constitutes a unilateral exit dressed in counterparty clothing, is unsettled. A sub-seller bonded under a CancellableSeller wrapper has consented to the wrapper’s terms at commit time, but consent at commit time to a future programmatic acknowledgement is structurally different from consent at the moment of cancellation; the design-discipline reading depends on which framing one accepts.

Cargo-handoff operationalization. A fresh transport process for the remaining legs requires the original carrier to release cargo to the new carrier, which is a real-world handoff with no described attestation surface in the present sketch.

Cancellation-fee arithmetic. Setting the fee too low produces an early-exit option the original sub-sellers will refuse; setting it too high forecloses the transfer use case. The present paper does

not defend any specific fee schedule; the arithmetic is left for the deployment-specific design.

The pattern is therefore not a recommended deployment artifact in the way the schemas of Section 6 are. We include the sketch because the BoL-transferability question is the load-bearing concern for any container-shipping deployment, and because the conventional answer (negotiable bills of lading endorsed mid-transit) is precisely what the kernel’s buyer-fixed invariant rules out. Deployments requiring BoL transferability can rely on conventional BoL doctrine through the off-chain agreement, or constrain their use cases to the non-transferable region the kernel admits directly.

Why MLETR-style transferable-record analysis applies. The MLETR analysis distinguishes records emitted by the kernel itself — commitment digests, bond deposits, the cumulative-value accumulator state, resolution events — from discretionary schema-validated attestations such as `container-seal-v1` events or `handoff-v1` attestations. The kernel byproducts are schema-typed at the kernel level and cryptographically authenticated; the MLETR Article 10–11 analysis applies to them directly, with the kernel’s structural properties (singularity through the cumulative-value accumulator, control through buyer-dominance, integrity through the chain’s consensus) discharging the control and singularity tests for transferable records. The discretionary schema-validated attestations are evidentiary artifacts *about* the underlying transferable record (the commitment), not separate transferable records inheriting MLETR status in their own right. The cancellable-seller / counter-process pattern preserves the kernel-byproduct fit because the transfer is achieved through bilateral signing on two separate processes rather than through unilateral endorsement of a single record; the protocol-layer transfer pattern does not violate either MLETR test.

What the bonded composition does and does not preserve. A useful organizing distinction is between order-form (named consignee, transferred by endorsement to a designated party) and bearer-form (transferred by physical delivery of the instrument without endorsement) bills of lading. Order-form endorsement is partially expressible through the cancellable-seller pattern above (subject to the underspecification axes). Bearer-form BoLs are functionally equivalent to a transfer-without-counterparty-consent pattern that the bonded composition does not support. Bearer-instrument practices remain common in commodity trading (oil cargoes, grain shipments) where the bill is treated as a near-money instrument [Goode and Gullifer, 2017]; non-bearer order-form practice predominates in container shipping but is not the only operational pattern. The bearer-form exclusion is an honest exclusion: parties whose operations depend on bearer-instrument liquidity should plan around the conventional regime for that part of their workflow.

8 Comparison with Existing Approaches

The architecture should be located against the existing landscape of digital-shipping infrastructures.

TradeLens. TradeLens was a permissioned consortium ledger; the architecture proposed here is a permissionless bonded settlement primitive. The functional comparison is asymmetric: TradeLens supplied information transparency without settlement; the bonded architecture supplies settlement (with information transparency emerging as a byproduct of the bonded record). The governance comparison is also asymmetric: TradeLens was governed by IBM–Maersk; the bonded architecture has no governance at the kernel layer. The carriers who declined TradeLens did so for a governance reason; the architecture proposed here removes the reason.

CargoX. CargoX is a blockchain-based bill-of-lading platform that emphasizes transferable-electronic-record functionality and MLETR compliance [CargoX, 2023]. CargoX is a focused product addressing the BoL transferability problem; the bonded architecture is a more general settlement primitive whose relationship to CargoX is composable rather than competitive. A shipper using the bonded architecture for the bilateral transport coordination could compose CargoX (or its successor) at the BoL layer, with the bonded record providing the underlying settlement evidence and the CargoX layer providing the transferable-instrument semantics. The two are not in competition; they address adjacent layers of the stack.

TradeTrust. TradeTrust, a Singapore-government-backed framework for electronic transferable records, supplies open-source tools and standards for issuing and managing electronic BoLs and similar instruments [TradeTrust, 2023]. Like CargoX, it is composable with the bonded architecture rather than competitive with it. TradeTrust’s standards-based approach to MLETR fit produces artifacts that are interoperable with the bonded composition’s evidence record.

MLETR. The UNCITRAL Model Law on Electronic Transferable Records (2017) is a legal framework rather than a technology platform [UNCITRAL, 2017]. The kernel-layer artifacts satisfy Articles 10 and 11 (functional equivalence, control, singularity) for non-bearer transferable records, with bearer-instrument semantics noted as an honest exclusion (Section 7).

The architectural distinction. The TradeLens replacement is not a better TradeLens. It is a re-architecture of the underlying coordination function from *shared ledger maintained by consortium* to *bonded settlement primitive composed by participants*. The shift is structurally substantial: the consortium’s role is absorbed into the kernel’s permissionless protocol; the carriers’ participation decision is reduced to whether to compose with the protocol on individual shipments rather than whether to ratify a competitor’s gatekeeping. The carriers who refused to ratify TradeLens are not asked to ratify anything in the bonded architecture; they participate by signing commitments to specific shipments and decline by not signing. The decision is made at the level of individual commerce rather than at the level of industry governance.

9 Scope

The argument bounds itself by several scope conditions.

Asset specificity at the port layer. Major ports are functional monopolies for the slot, terminal capacity, and operational infrastructure they provide; carriers have limited alternative-supplier leverage at congested hubs. The architecture’s competitive properties at the port-supply layer are constrained accordingly: where the port-of-call is dictated by the route and the port operator is a near-monopolist, the buyer-dominance leverage at the sub-procurement is attenuated. The architecture continues to function in these cases (the bond posture is well-defined; the resolution path is clean) but it does not produce price competition in port services where structural conditions don’t admit it.

Customs and sovereign authorities. Customs authorities have two distinct functions in this composition. As bonded sellers of border-clearance service (charging the importer for handling, examination capacity, filing-receipt processing, etc.) they are RWA-as-wallet on the same footing as commercial providers, and their participation in the bonded process is unproblematic at the kernel layer. As sovereign decisional authorities on whether specific goods clear under the relevant tariff schedule, they continue to operate in their own register, off-chain; the architecture does not re-engineer customs administration and the kernel does not encode the sovereign decision. The bonded composition produces a verifiable evidence record on which customs decisions

can operate and which customs decisions are recorded against; this is an improvement over the paper-and-database baseline but not a substitute for the sovereign authority itself.

Bond-currency at the small-shipper end. Small shippers without crypto-treasury infrastructure face operational friction at the bonding step. The constraint is real but solvable through standard treasury-of-record arrangements (a forwarder bonds on behalf of small shippers, with the small shipper paying conventional fees off-chain to the forwarder for the bonding service). This is a runtime-tier implementation question rather than a substrate-tier question; the substrate is denomination-agnostic.

Insurance and reinsurance. Marine insurance, P&I clubs, and the general apparatus of shipping-risk underwriting continue to operate. The bonded architecture changes the risk profile that the marine-insurance markets price (smaller residual after bond allocation; concentrated risk at the carrier-bond layer; a new class of bond-default risk at the port-supplier layer) but does not displace the insurance function. We expect the marine-insurance markets to develop fluent products around the bonded architecture as the architecture diffuses.

Network effects and adoption. The architecture’s adoption dynamics differ from TradeLens’s in that no participant must ratify any other participant’s governance position. A shipper, forwarder, or carrier can participate on individual shipments without joining anything; the participation decision is shipment-by-shipment rather than platform-membership-shaped. We do not model the diffusion trajectory beyond noting that the structural barrier that defeated TradeLens (consortium ratification by competitors) is absent here.

10 Conclusion

TradeLens failed for governance reasons that no consortium-governance fix can resolve, and the structural alternative is permissionless, ownerless infrastructure at the platform layer rather than a better consortium. The bonded primitive in **FigaroCore** supplies the architectural shape: each leg of a shipment is a bilateral bonded commitment; progressive collateralization scales the primitive across the multi-leg fan-out; buyer dominance plus atomic resolution propagates damage to the unit causing the slip; the schemas the composition reuses are largely existing infrastructure with two well-bounded additions.

The composition operates at the same perimeter TradeLens attempted — inter-logistics coordination among carriers, ports, customs authorities, freight forwarders, NVOCCs, customs agents, trade-finance parties, marine insurers, and inspection services — with the buyer-of-record at the relevant tier as the **rootBuyer**. Each provider is an RWA-as-wallet whose participation in this market is sustained by receipts covering the asset’s off-chain operating expenses; public-authority wallets participate on the same footing as commercial. Granularity within the perimeter is a kernel-blind design choice; the fine-grained version developed here makes the value-adder structure explicit, and coarser aggregations are admissible where operational arrangements support them.

What the present paper supplies is the composition — the organizational DAG, the per-edge bonding mechanics, the schema set including the two new schemas (**container-seal-v1**, **incoterms-2020-v1**), and the cancellable-seller / counter-process pattern for BoL transferability — that turns the bonded primitive into a working inter-logistics coordination architecture at the same perimeter TradeLens attempted and could not sustain.

Acknowledgments

The TradeLens worked example originated in a project repository of agent-coordination scenarios. The present paper formalises the scenario into an industrial-engineering register.

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